

進階機器學習

Homework #4

Due 2025 **April 25** 11:00 pm

1. Utilize the CLIP model to perform image classification on the dataset provided in ***HW2_MNIST_train.zip***. Evaluate and report the Top-1 and Top-3 classification accuracy.
2. Split the dataset in ***HW3_text.csv*** into 70% training and 30% testing subsets. Fine-tune a BERT model using the training data and perform text classification on the test set. Present the confusion matrix based on the test results.
3. Demonstrate the derivation of the total number of parameters (weights) in the LLaMA2-7B model. (Note: LLaMA2-7B uses the grouped-query attention mechanism—please take this into account in your calculation.)
4. Implement a simple question-answering (QA) system using the Chroma vector database. Provide a brief demonstration of its functionality.
5. Develop a Python program that accepts an image and a textual description, and determines whether the content of the image matches the description. Report whether the match is valid or not.